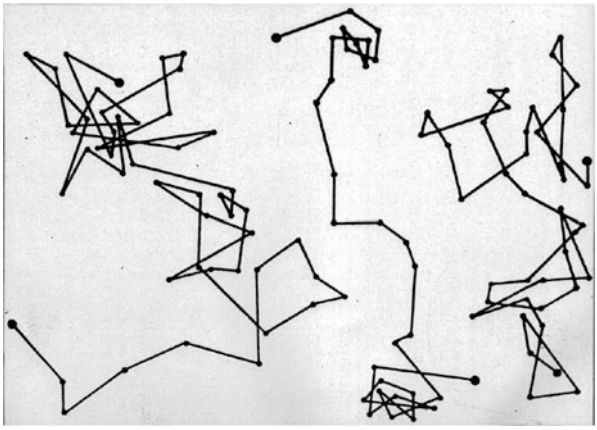


Fig. 6.3 Perrin's plot of the motion of small particles formed from resin of the tree *garcinia xanthochymus* buffeted by interactions with a bath of water molecules. Perrin's careful observations were consistent with those of earlier observers, beginning with Brown's study of pollen grains in 1827

were worked out in the late 1800s by Thorvald Thiele and Louis Bachelier, although the complete picture was only obtained in the early 1900s in celebrated works by Albert Einstein and Marian Smoluchowski. Brownian motion describes the process by which particles placed in a fluid bath move about subject to collisions with the molecules of the fluid. These collisions, which for all practical purposes may be viewed as random, introduce a jerky character to motion such as that shown in Fig. 6.3.

To model the randomized motion of the Brownian particle, the concept of a random walk is typically used. A random walk is an example of a stochastic process, a collection of random variables parameterized by either a discrete or continuous index parameter [269, 314]. A random walk is a discrete stochastic process in which the state X_n at a given instant (defined by the index n) is related to the state X_{n-1}, at step $n-1$ by an offset that may be viewed as a random variable. That is, we have

$$\mathrm{X}_{n+1} = \mathrm{X}_n + \delta x \mathrm{J}_n.$$

We initiate the random walk with $\mathrm{X}_0 = 0$. The quantity J_n is a random variable drawn from a certain probability distribution. In the classic model of a random walk the jump takes values ± 1 with equal probability. The *increments* of the random walk are the differences $\Delta \mathrm{X}_n = \mathrm{X}_n - \mathrm{X}_{n-1} = \delta x \mathrm{J}_{n-1}$, thus (X_n) is a random process with independent increments.

The solution X_n of the random walk may be written

$$\mathrm{X}_n = \delta x \sum_{k=0}^{n-1} \mathrm{J}_k,$$

hence the variance is

$$\mathbb{E}\mathrm{X}_n^2 = \delta x^2 \mathbb{E}\left(\sum_{k=0}^{n-1} \mathrm{J}_k \right)^2.$$

Expanding the squared sum we see that there are (i) n squares J_k^2 plus (ii) products of the form $\mathrm{J}_k\mathrm{J}_l$, $k \neq l$. Clearly the former are always one and the latter have zero expectation, thus

$$\mathbb{E}\mathrm{X}_n^2 = n\delta x^2.$$

The variance therefore grows linearly with the number of steps taken.

Now let us take a slightly different perspective. We start with a time interval $[0, \tau]$ and subdivide this into ν steps of size $\delta t = \tau/\nu$. Let $t_n = n\delta t$, $n = 0, 1, \ldots, \nu$ and define

$$\mathrm{Y}(t_{k+1}) = \mathrm{Y}(t_k) + \sqrt{\delta t}\mathrm{J}_k, \qquad k = 0, 1, \ldots, \nu - 1,$$

where $\mathrm{Y}(0) = 0$. Effectively we are making the size of the jump δx taken at each step of the random walk equal to $\sqrt{\delta t}$.

We will use the shorthand $\mathrm{Y}_k = \mathrm{Y}(t_k)$ and $\Delta\mathrm{Y}_k = \mathrm{Y}_{k+1} - \mathrm{Y}_k = \sqrt{\delta t}\mathrm{J}_k$.

The increments are clearly independent. Let us observe the following facts, for $0 \leq k \leq l \leq \nu$

(i) $\mathbb{E}(\mathrm{Y}_l - \mathrm{Y}_k) = 0$,
(ii) $\mathrm{Var}(\mathrm{Y}_l - \mathrm{Y}_k) = t_l - t_k$.

For points between t_k and t_{k+1} we define the value of $\mathrm{Y}(t)$ by linear interpolation:

$$\mathrm{Y}(t) = \mathrm{Y}_k + \frac{(t - t_k)}{\delta t}\Delta\mathrm{Y}_k$$

Then, for $0 \leq t \leq s \leq \tau$, we have

$$\mathbb{E}(\mathrm{Y}(s) - \mathrm{Y}(t)) = 0.$$

This is easy to demonstrate. However, the intuitive corresponding statement $\mathrm{Var}(\mathrm{Y}(s) - \mathrm{Y}(t)) = (s - t)$ does not quite hold due to the reliance on interpolation. Nevertheless, such a relation holds in an approximate sense and in the limit of small step.

The relation $\delta t = \delta x^2$ defines a scaling of time in relation to distance. When we take limits as these two quantities go to zero in such a way that $\delta t = \delta x^2$ we refer to it as a *diffusion limit*. It is this type of random diffusion model that has the greatest relevance for experiments like those of Brown and Perrin.

In the diffusion limit the motion of the random walk can be viewed as an everywhere continuous (but almost nowhere smooth) path which is called a *Wiener process* (after Norbert Wiener).

$$\mathrm{W}(t) = \widehat{\lim}_{\delta t \to 0} \sqrt{\delta t} \sum_{k=0}^{\lfloor t/\delta t \rfloor} \mathrm{J}_k.$$